06/01/2026 (FN)

Hall Ticket Number:

MR-21

Code: B315CB6S2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, December 2025/January 2026

Computer Organization and Architecture

(Common to CSB, CSM and CSD)

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Write about hierarchy of buses, bus signals and its functionalities.	2	1	1	1
1. b)	Briefly explain phases of instruction cycle	2	2	1	2
1. c)	Distinguish between High level and low level languages.	2	4	2	2
1. d)	What is an Interrupt?	2	2	2	1
1. e)	Draw block diagram for a bidirectional input/output device with its interface and enable/load logic?	2	2	3	1
1. f)	List out the Register transfer notations for Arithmetic Micro Operations.	2	4	3	4
1. g)	List out the importance of interfacing.	2	1	4	2
1. h)	List out the methodologies involved in handling Priority Interrupt.	2	1	4	3
1. i)	Write about pipelining and its importance in high speed applications.	2	2	5	1
1. j)	List out the general aspects of ROM, RAM and IO interfacing modules.	2	2	5	2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the basic components of Generic Computer system regardless of its internal architecture.	5	2	1	1
2. b)	Sketch the general register organization of CPU with its functionalities and block diagram.	5	2	1	1
	OR				
2. c)	Elaborate how CPU is concordant with its Input & Output devices and explain the interfacing modules involved?	5	2	4	3
2. d)	With the help of examples explain logical micro operations.	5	2	1	1
3. a)	With a neat schematic, explain the steps involved in fetch and decode phases using register transfer instructions.	5	2	1	2
3. b)	With a neat flowchart, explain how the control unit determines the instruction after decoding an instruction.	5	3	1	1
	OR				
3. c)	Implement 4-bit Binary Adder-Subtractor and Binary Incrementor.	5	3	2	4
3. d)	Explain the logical micro operations which manipulates individual bits of word in register with examples.	5	4	2	1
4. a)	Explain about address sequencing in control memory with neat diagrams?	5	2	3	1
4. b)	Apply 2's compliment method to perform subration between two given binary numbers 1010 1110 – 1100 0011. Elaborate the steps involved.	5	3	3	3
	OR				
4. c)	Design a 4-bit ALU which performs arithmetic, Logical and shift operations.	5	2	3	3
4. d)	Write about hardware organization of micro programmed control unit.	5	2	3	1

P.T.O.

5. a)	Discuss the Memory Hierarchy in computer system with regard to Speed, Size and Cost?	5	3	4	2
5. b)	Explain the mechanism involved in Magnetic Disks and Magnetic Tapes.	5	5	4	1
	OR				
5. c)	Explain about DMA controller and its mode of data transfer.	5	2	4	1
5. d)	Contrast Daisy-Chaining priority and Parallel priority Interrupt with its hardware diagram.	5	4	4	1
6. a)	Explain four segment CPU pipeline and Timing of instruction pipeline.	5	2	5	1
6. b)	Write about pipelining and its importance in high speed applications.	5	2	5	2
	OR				
6. c)	Explain inter processor communication and synchronization in a shared multiprocessor environment.	5	5	5	2
6. d)	Illustrate serial and parallel arbitration produces in a shared multiprocessor environment.	5	3	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome